CATHOLIC JUNIOR COLLEGE
JC2 Preliminary Examinations
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME				
CLASS	2T			

INDEX NUMBER				

BIOLOGY 9648/02

Paper 2 Core Paper

28 August 2015

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the question paper.

Section B
Answer any **one** question on the answer paper provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units

At the end of the examination, fasten all work securely together.
The number of marks is given in brackets [ ] at the end of each question or part of the question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
Total	

This document consists of **24** printed pages.

Section A

Answer **all** the questions in this section.

1 Meiosis is an important event which is required) prior to fertilisation in sexual reproduction.

(a) Explain the need for reduction division (meiosis) prior to fertilisation in sexual reproduction. [2]

- Meiosis halves the number of chromosomes going into gametes ;
- so that the diploid number in the zygote can be conserved when the gametes fused together during fertilisation ;

OR

- During fertilisation, the number of chromosomes doubles in the zygote ;
- In order to conserve the diploid number in the organism, there has to be an equivalent reduction in the chromosome number in the gametes that fused together ;

Fig. 2.1 below shows 8 different micrographs taken at the different timing during the meiosis of a cell.

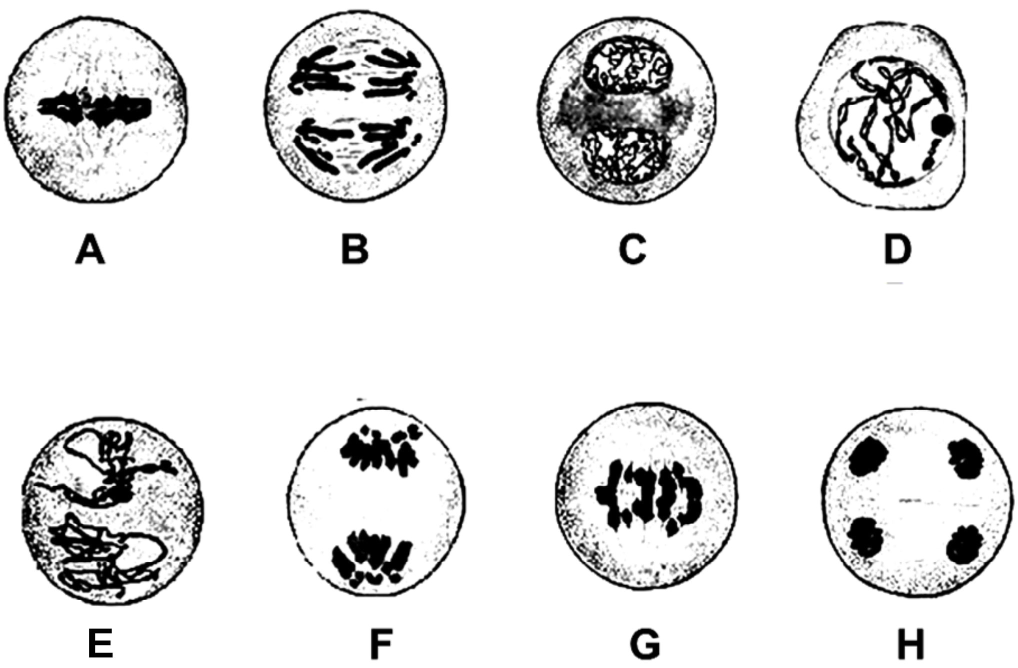


Fig. 2.1

(b) Arrange the 8 stages (A - H) in chronological order. [1]

D, **A**, **G**, **F**, **C**, **E**, **B**, **H**

(c) With reference to Fig. 2.1,

(i) identify **one** stage (A - H) that will result in genetic variation. [1]

(ii) explain how this stage results in genetic variation. [2]

(i) Micrograph B ;
(ii) chiasmata formation and crossing over during prophase I resulting in the exchange of alleles ;
This results in new combinations of alleles on chromosomes of the gametes, causing genetic variation ;

OR

(i) Micrograph A ;
(ii) independent assortment of homologous chromosomes at metaphase I ;
Subsequent separation of the homologous chromosomes during anaphase I will produce different combinations of chromosomes in the daughter cells at the end of meiosis I.

OR

(i) Micrograph G;
(ii) independent assortment of chromatids at metaphase II ;
Subsequent separation of chromatids during anaphase II will produce different combinations of chromosomes in the daughter cells at the end of meiosis II ;

Bone marrow contains stem cells that divide by mitosis to form blood cells. Each time a stem cell divides it forms a replacement stem cell and a cell that develops into a blood cell.

Fig. 3.1 shows changes in the mass of DNA in a human stem cell from the bone marrow during three cell cycles.

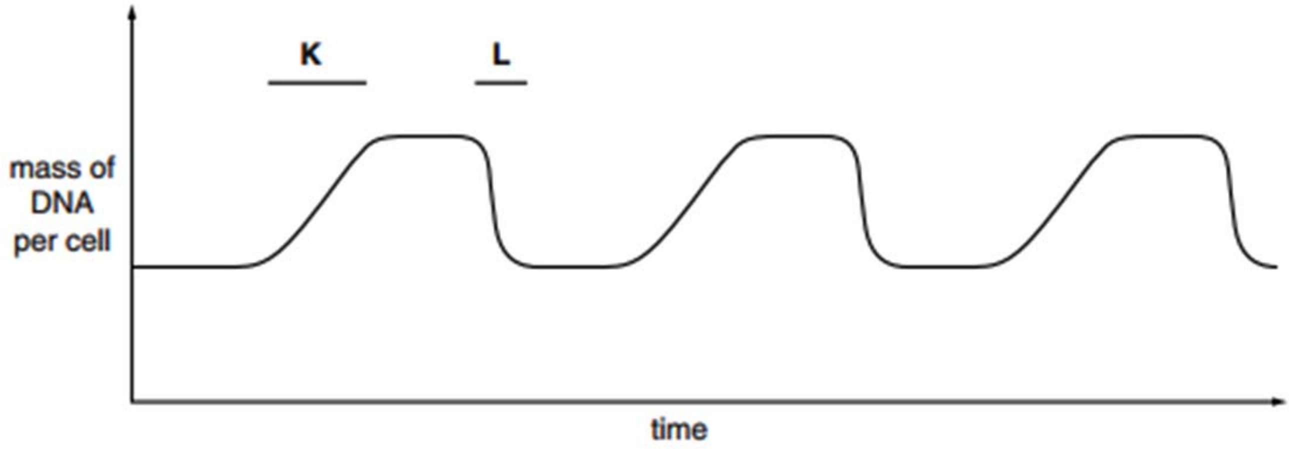


Fig. 3.1

(c) With reference to Fig. 3.1, state:

(i) what happens to bring about the changes in the mass of DNA per cell at **K** and at **L**. [2]

K – (DNA) replication/ synthesis / described ;
L – cytokinesis / cytoplasmic division / cell division ;

(ii) how many blood cells are formed from the stem cell in the time shown. [1]
3

(iii) what happens to the number of chromosomes in the stem cell. [1]
remain the same / stays constant / stay at 46 / AW ; *Reject: description of events occurring before and during mitosis*

[Total: 10]

2 Phosphate ions have a number of uses in organisms. These include:

- involvement in cell signalling responses
- involvement in energy transfer processes
- component of phospholipids
- component of nucleotides.

Phosphatase enzymes remove phosphate groups from organic compounds, while kinase enzymes add phosphate groups.

(a) Read the following passage:

Some hormones circulating in the blood are able to trigger transcription within a cell, even though they are unable to enter the cell. Phosphatases and kinases then take part in cell activities that eventually result in genes switching on and transcription beginning.

(i) Suggest why the hormones, referred to in the passage, are unable to enter the cell. [2]

- protein/peptide hormones ;
- too large to cross membrane ;
- hydrophilic/water soluble ; Accept: not, hydrophobic/lipid soluble
- unable to pass through hydrophobic core / AW of phospholipid bilayer ; [max 2]

(ii) Use the information in the passage to outline the process of cell signalling. [3]

- chemicals released are circulating hormones ;
- hormones combine with cell surface receptors ;
- on target cells / cells where transcription is triggered ;
- action of kinases and phosphatases (within the cell) lead to (specific) response ;
- specific response = transcription / production of mRNA ; [max 3]

(b) The activity of a phosphatase enzyme was measured at different values of pH by using nine different buffer solutions. The temperature was kept constant at 30 ^{o}C.

The results are shown in Fig. 4.1.

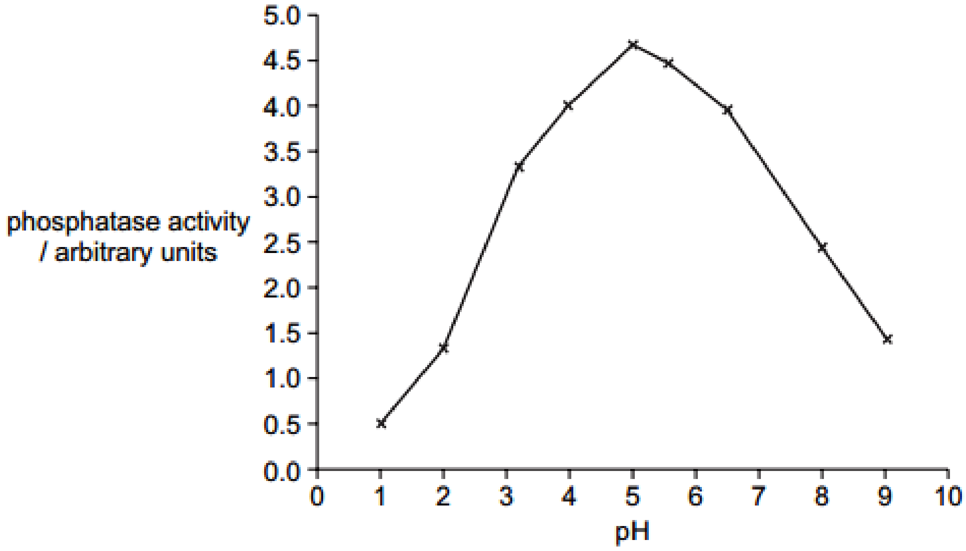


Fig. 4.1

(i) With reference to Fig. 4.1, describe the effect of pH on the activity of phosphatase. [2]

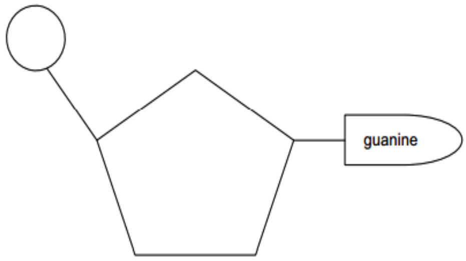
- optimum is, pH 5 / between pH 4–5.5 ; **A** optimum pH value between 4–5
- increasing activity as pH increases to, optimum / pH 5 ;
- decreasing activity as pH increases above, optimum / pH 5 ;
- active, over a wide pH range / between pH 1–9 ; [max 2]

(ii) Explain why the activity of phosphatase at pH 1 is very low. [3]

- low pH equivalent to high, hydrogen ion / H+, concentration ;
- hydrogen / ionic, bonds, disrupted / broken / AW ;
- active site shape, changed / AW ; **A** active site no longer complementary to substrate ref. to partial denaturation / some enzymes denatured ; (active site change so) decreases effective collisions / fewer enzyme substrate complexes formed ;
- (only) some (phosphatase) enzymes active / all enzymes partly active ; [max 3]

(c) Phosphatases are being used commercially to produce dephosphorylated DNA. This involves removal of the two terminal phosphate groups using phosphatases.

(i) In the box below, draw a labelled diagram of a dephosphorylated DNA nucleotide containing guanine base. [1]



(ii) Compare the structure of an ATP molecule with the DNA nucleotide drawn in **(c)(i)**.[4]

similarities
both have, pentose / 5C sugar ;
both have, organic / nitrogenous, base ; Accept: both have purine (base)
both have phosphate ;

differences
(ATP) ribose not deoxyribose ;
(ATP) adenine not guanine ;
(ATP) three phosphates, not one ;

[Total: 15]

3 CFTR gene is controlled by a promoter that has a variable proximal region. Detailed investigation reveals that gene expression is dependent on the conservation of a particular sequence within the proximal region. Table 3.1 shows the results of the investigation.

Table 3.1

Individual	Proximal region of the promoter sequence	Binding of RNA polymerase II	Expression of CFTR
1	5' – AATTGGAAGCAAAT – 3'	Able to bind	Basal level
2	5' – CCTTGGAAGCAAAT – 3'	Able to bind	Basal level
3	5' – TTTTGGAAGCAAAT – 3'	Able to bind	Basal level
4	5' – AATTAAGGTCAAAT – 3'	Able to bind	Basal level
5	5' – AATTAAGGTCATTT – 3'	Unable to bind	Absent
6	5' – AATTAAGGTTGAAC – 3'	Unable to bind	Absent
7	5' – AATTGGAAGCAAAC – 3'	Unable to bind	Absent

(a) With reference to Table 3.1;

(i) State the conserved sequence that must present for the CFTR gene to be expressed. [1]

5' – CAAAT – 3' ;

(ii) Explain why gene expression was absent in individual 5. [2]

- Contains sequence CATTT instead of conserved sequence ;
- Promoter not complementary in shape to RNA Polymerase II ;
- Transcription initiation complex not formed ;

Gene mutation responsible for cystic fibrosis can be classified into 6 different classes.

Class I and **II** CFTR mutations are typically characterised by a reduction in the quantity of functional CFTR protein.

- **Class I** mutations can result from nonsense and frame-shift mutations, as well as mRNA splicing defects.
- **Class II** mutations are caused by folding or maturation defects, which can result in premature CFTR degradation.

Class III and **IV** mutations, however, are typified by aberrant channel function rather than reduced quantities of CFTR.

- CFTR channels with **Class III** mutations result in limited channel gating that arises from ineffectual binding of nucleotide.
- CFTR channels with **Class IV** mutations are able to open and close; however, chloride and bicarbonate ions are unable to freely pass through the channel due to conductance defects.

While **Class V** mutations can lead to the production of normal CFTR, a limitation of transcriptional regulation results in a reduced quantity of the protein being produced.

Finally, the relatively novel **Class VI** mutations are characterised by high turnover of CFTR at the cell surface.

(c) Explain how **Class I** mutations result in a reduction in the quantity of functional CFTR protein. [2]

- Nonsense mutation results in a truncated protein being formed, which will not be able to fold into a functional protein ;
- Frame-shift mutation results in a change in amino acid sequence, resulting in a non-functional protein being formed ;
- mRNA splicing defects will not result in a mature mRNA formed/does not have the correct continuous exon sequence, hence resulting in a non-functional protein being formed ; [max 2 marks]

(d) Suggest how Class V mutations result in a reduced quantity of protein formed. [2]

Mutation at enhancer region; – increased affinity for activator.
Mutation at silencer; – increased affinity for repressor protein.
Mutation at promoter; – reduced affinity for RNA Polymerase.
Mutation at gene coding for activator protein; – decreased affinity for enhancer region.
Mutation at gene coding for repressor protein; – increased affinity for silencer region.
Mutation at gene coding for RNA Polymerase; – increased affinity for promoter region.

(e) Explain why a high turnover of CFTR at the cell surface results in cystic fibrosis. [2]

- **CFTR proteins** are being **removed** from the cell surface membrane **at a higher rate**.
- This results in the **decrease number of functional CFTR proteins on the cell surface** to regulate the movement of chloride ions.

[Total: 9]

4 Fig. 4.1 shows how Histone 3 in eukaryotes may be modified for gene expression. Phosphorylation of Histone 3 (H3) at position S10 results in rapid acetylation, which in turn leads to increased gene expression.

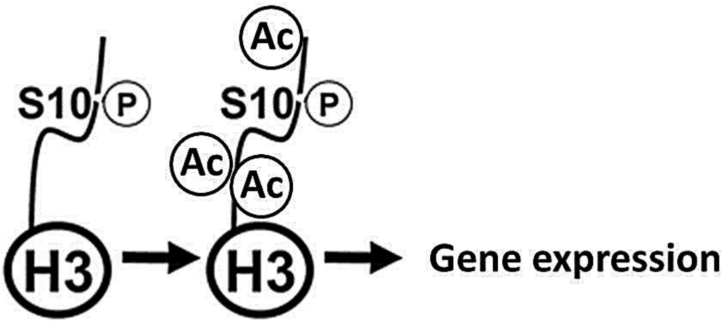


Fig. 4.1

(a) Suggest how the phosphorylation of H3 would result in the rapid acetylation of the protein. [2]

- The phosphate groups present on H3 facilitates the binding/docking of Histone acetyltransferase (HAT) onto the histone ;
- With the stable binding of HAT, the enzyme can proceed to catalyse the acetylation process ;

(b) Explain how the acetylation of H3 leads to increased gene expression. [4]

- The addition of acetylation onto lysine residues on the histone tails will neutralise the negative charges that these residues carry ;
- As a result, the electrostatic interaction between the histone proteins and the negatively-charged DNA will be lost ;
- This would cause chromatin to be less compactly packed / more loosely packed ;
- Transcription factors can more easily access the promoter and bind to it to initiate transcription ;

[Total: 6]

5 MRSA is a variety of *Staphylococcus aureus*. It is difficult to treat infections caused by this bacterium because it is resistant to methicillin and to some other antibiotics. As a result, some patients who are already very ill may die if they become infected with MRSA.

Fig. 5.1 shows the number of deaths in England and Wales between 1994 and 2008 caused by MRSA.

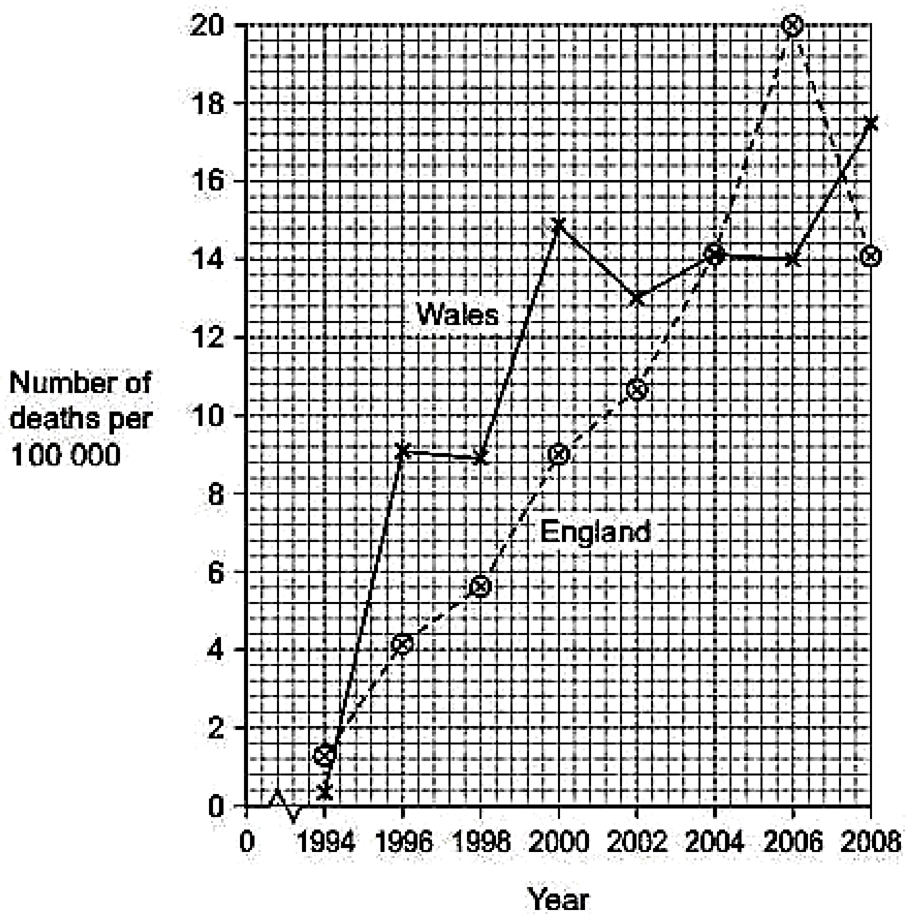


Fig. 5.1

(a) The antibiotic methicillin inhibits the enzyme transpeptidase. This enzyme is used by some bacteria to join monomers together during cell wall formation. Methicillin has a similar structure to these monomers.

(i) Use this information to explain how methicillin inhibits the enzyme transpeptidase. [2]

1. Attaches to active site (of enzyme)/Competes for active site of enzyme ;
2. (Methicillin) is a competitive inhibitor / prevents monomers/substrate attaching (to enzyme) ;

(ii) Describe the change in the number of deaths caused by MRSA in England in the period shown in the graph. [1]

1. Increase up to 2006/20 (per 100 000) then decreases till 2008/14 (per 100 000) ;

(iii) Calculate the percentage increase in the number of deaths caused by MRSA in Wales from 1996 to 2006. Show your working. [1]

Correct answer in range of 52 – 59.1% = two marks ;

Incorrect answer but shows change as between 4.8 – 5.2 / shows correct subtraction giving this change e.g. 14 - 9 = one mark ;

(b) Describe how gene transmission and selection have increased the difficulty of treating bacterial infections with antibiotics. [6]

(Antibiotic) resistant gene/allele ;
Vertical (gene) transmission (Binary Fission) ;
Resistant bacteria (survive and) reproduce / population of resistant bacteria increases ;
Increase in frequency of (resistant) allele/gene (in future generations) ;
Horizontal (gene) transmission ;
Plasmid ;
Conjugation / pilus (tube); Transformation ;
(Horizontal transmission/ conjugation) can occur between bacteria of different species ;

[Total: 10]

6 In mice, fur colour is controlled by a gene with multiple alleles. These alleles are listed below in no particular order.

yellow = C^y agouti = C^a black = C^b

(a) Suggest explanations for the results of the following crosses between mice.

(i) Mice with agouti fur crossed with mice with black fur may produce all agouti offspring or some agouti and some black offspring. [2]

accept answers in a genetic diagram where genotypes are linked to phenotypes

- agouti allele / C^a , dominant to black allele / C^b ; (or reverse argument)
- black parents homozygous recessive ;
- agouti parents heterozygous or homozygous ; [max 2 marks]

(ii) Crosses between heterozygous parents with the genotype C^y C^b always produce a ratio of two yellow mice to one black mouse. [2]

accept answers in a genetic diagram where genotypes are linked to phenotypes

- yellow allele / C^y , dominant to, black allele / C^b ;
- ref. to modified 3:1 ;
- (homozygous) genotype C^y C^y , lethal / does not survive ; [max 2 marks]

(b) The budgerigar, *Melopsittacus undulatus*, is a small type of parrot that is native to Australia.

Fig. 6.1 shows a budgerigar.

A budgerigar can have blue, green, yellow or white feathers.

Two genes, **A/a** and **D/d**, are involved in the inheritance of feather colour in budgerigars.

- A bird which has at least one dominant allele **A** but is homozygous for **d** has blue feathers.
- A bird which has at least one dominant allele **D** but is homozygous for **a** has yellow feathers.
- A bird with at least one dominant **A** allele and one dominant **D** allele has green feathers.
- A bird that is homozygous for **a** and **d** has white feathers.

Two green-feathered budgerigars, heterozygous at both gene loci, were crossed. Draw a genetic diagram of this cross to show the probability of producing offspring with yellow feathers. [6]

- parental genotypes AaDd × AaDd ;
- gametes AD Ad aD ad × AD Ad aD ad ;
- two marks for correct Punnett square ;; deduct one mark for each mistake
- two marks for correct Punnett square ;; deduct one mark for each mistake
- (all 4) phenotypes linked correctly to genotypes ;
- (probability of yellow offspring) 3 out of 16 or 0.19 or 19% ;

[Total: 10]

7 Fig. 7.1 shows a neurone forming three synapses with adjacent neurones.

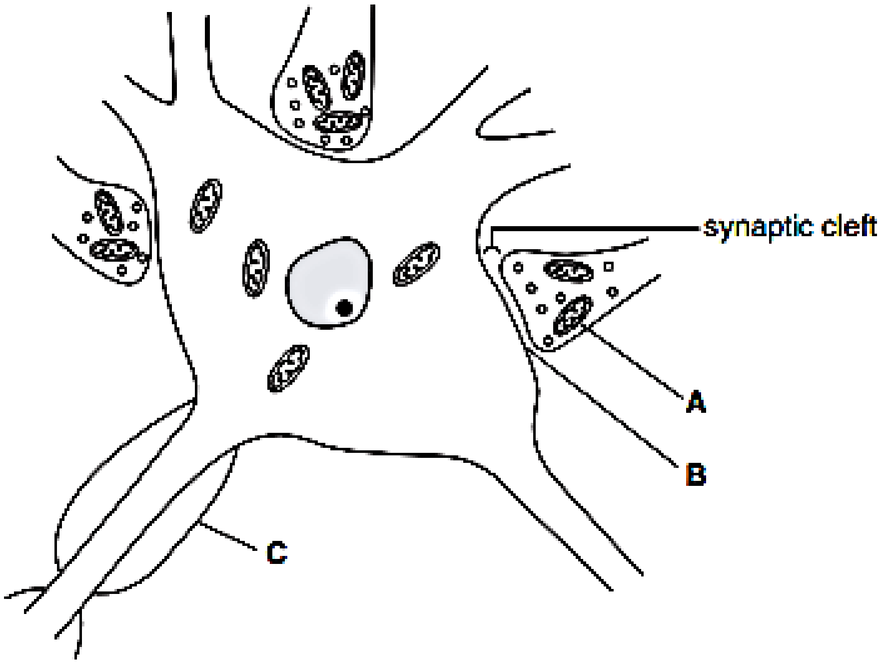


Fig. 7.1

(a) Name **A**, **B** and **C**. [3]

A – mitochondrion ;
B – post-synaptic membrane ;
C – myelin sheath / Schwann cell ;

(b) State how synapses ensure one-way transmission of nerve impulses. [1]

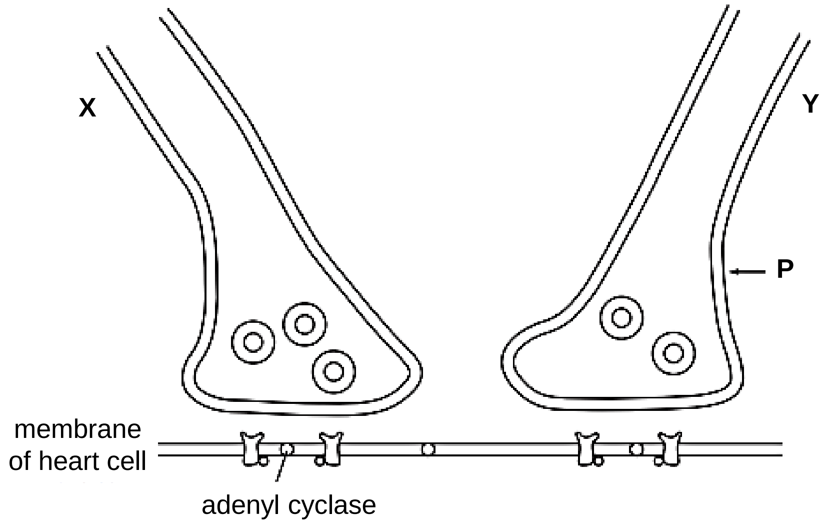
1. vesicles found only in presynaptic neurone / ACh released only from presynaptic neurone or membrane ;
2. receptor (proteins) found only on postsynaptic membrane ; [max 1 mark]

(c) In a learning activity, it is believed that the number of synapses between brain neurones increases.
Suggest the advantages of this increased number of synapses. [2]

1. allows more interconnection of nerve pathways / AW ;
2. allows wider range of responses ;
3. allows for summation of impulses from multiple neurons ;
4. AVP ;

(d) Fig. 7.2 shows endings of neurones **X** and **Y** in the heart.

Stimulation of **Y** leads to an increase in the concentration of the second messenger, cyclic AMP (cAMP), in the cytoplasm of the heart cell and the stimulation of **X** leads to a decrease.



Describe the events that follow the arrival of an impulse at **P** on neurone **Y** that leads to an increase in cAMP. [4]

voltage-gated (calcium) ion channels open ;
calcium ions diffuse into the, (presynaptic) neurone / synaptic bulb ;
vesicles containing neurotransmitter move to presynaptic membrane ;
vesicles fuse with membrane / exocytosis ;
neurotransmitter diffuses across, synapse / gap ;
binds to receptor ;
activates G protein ;
adenyl cyclase catalyses formation of cyclic AMP ;
from ATP ; [max 4]

[Total: 10]

8 (a) Fig. 8.1 show the location of two types of pistol shrimp with reference to the Panama strip.

Panama is a narrow strip of land which today joins North America and South America. It was formed by land moving up from beneath the sea. Panama has separated the Pacific Ocean and the Caribbean Sea for the past 3 million years.

Scientists put one **Type A** shrimp and one **Type B** shrimp together in a tank of seawater. The two types of shrimp snapped their claws aggressively at each other. They did not mate. The scientists said that this was evidence for the **Type A** and **Type B** shrimps being classified as two different species.

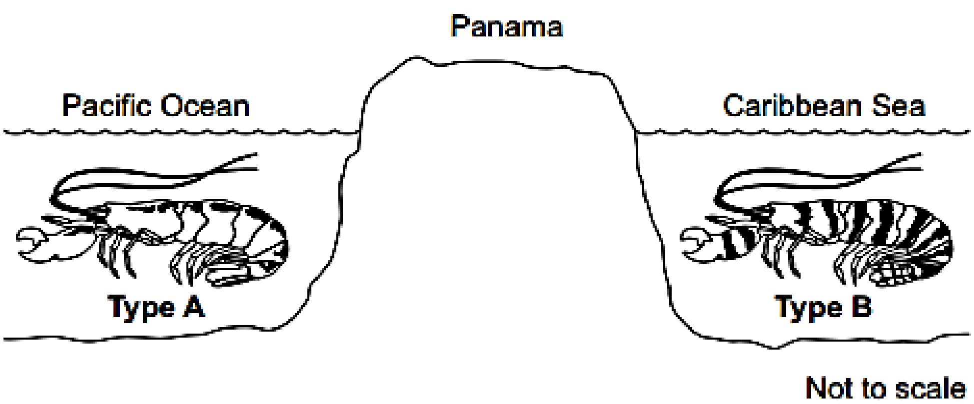


Fig. 8.1

With reference to the context given, explain how two different species of pistol shrimp could have developed from an ancestral species of shrimp. [5]

(two ancestral populations) separated (by geographical barrier / by land) / were isolated
genetic variation (in each population) or different / new alleles or mutations occur
different environment / conditions
natural selection occurs or some phenotypes survived or some genotypes survived,
(favourable) alleles / genes / mutations passed on (in each population)
eventually two types can no longer interbreed successfully/ can no longer produce fertile offspring

(b) Scientists used DNA hybridisation to determine the evolutionary relationships between five species of primate. The temperature at which a molecule of double-stranded DNA separates into two single strands is the separation temperature. The scientists recorded the mean separation temperature of DNA in which both strands were from the same species. The scientists then recorded the mean decrease in separation temperature of DNA in which one of the strands was from another species. Their results are shown in Table 6.

Table 6

Primate	**Mean decrease in separation temperature/ °C**				
	Human	Chimpanzee	Gorilla	Orang-utan	Gibbon
Human					
Chimpanzee	1.7				
Gorilla	2.3	2.3			
Orang-utan	3.6	3.6	3.5		
Gibbon	4.8	4.8	4.7	4.9	

(i) These data suggest that gibbons are the most distantly related to humans. Explain how. [2]

1. (So) few(er) hydrogen/H bonds ;
2. (So) few(er) complementary bases/ few(er) base pairs ;

(ii) There were differences in separation temperature of DNA formed from single-stranded DNA of the **same** species of primate. Suggest why. [1]

(Same species) have different alleles/different base sequences/ (different) mutations/introns/ non-coding DNA/multiple repeats;

(iii) The scientists assumed that the decreases in separation temperatures are directly proportional to the time since the evolutionary lines of these primates separated. Gorillas are thought to have separated from orang-utans 20 million years ago.

Use this information to calculate how long ago the evolutionary lines of humans and chimpanzees separated. Show your working. [2]

Correct answer in range of 9.69 to 9.71(4286) = 2 marks;; Accept: 9 690 000 to 9 714 286 for 2 marks

If 10 is shown and an answer in the range of 9.69 to 9.71(4286), award 2 marks
If 10 is shown and an answer in the range of 9.69 to 9.71(4286) is not shown, award 1 mark

One mark for incorrect answers that show any of the following:
(1 °C =) **5.7(14286)** (million years) OR: **20 000 000 ÷ 3.5** OR: **20 ÷ 3.5**

[Total: 10]

Section B

Answer **one** question.

Write your answers on separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in ***continuous prose***, where appropriate.
Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

9 **(a)** Describe the various types of genetic changes in proto-oncogenes which can result in cancer. [7]

- **Movement of DNA within the genome – chromosomal translocation** ;
- In cancer cells, **chromosomes** are commonly found to have **broken** and **rejoined incorrectly** and there is **translocating fragments** from **one chromosome to another** ;
- **Chromosomal translocation that fuses two genes together to produce a hybrid gene** encoding **a chimeric protein** (a fusion protein created from joining two or more genes originally coding for separate proteins) whose activity, unlike that of the parent proteins, often is **constitutive** ;
- If a translocated proto-oncogene ends up near **an especially active promoter** (or other control element), its transcription may increase, leading to an **increase in expression of normal growth stimulating protein resulting in an abnormal cell cycle**. Thus, the proto-oncogene converts into an oncogene.

- **Gene amplification caused by errors during DNA replication** ;
- **Gene amplification increases the number of copies of the proto-oncogene in the cell**, leading to an increase in its transcription resulting in **overproduction of the encoded protein.** This results in an excess of normal growth-stimulating protein ;
- **Point mutation in either the promoter or an enhancer** that controls a proto-oncogene, causing an **increase in its expression**.
- **Point mutation in the coding sequence of the proto-oncogene**, changing the gene's product to a protein that is **more active** or **more resistant** to degradation than the normal protein ; [max 7 marks]

(b) Explain, with named examples, how environmental factors act as forces of natural selection. [8]

Example 1 - Selection Pressure 1

Example 1 - Selective Advantage/Disadvantage

Example 1- Outcome of Natural Selection

Example 1- Outcome of Natural Selection

Example 2 - Selection Pressure 2

Example 2 - Selective Advantage/Disadvantage

Example 2- Outcome of Natural Selection

Example 2- Outcome of Natural Selection

In the example *Biston betularia*, the **selection pressure is predator**, the environmental factor was soot in the industrialized areas.

Biston betularia carbonica / the **dark form of the peppered moth were at a selective advantage after the industrial revolution** because they were **better camouflaged against the bark of trees covered in soot**

The **light form/ *Biston betularia typica*** who were at a **selective disadvantage** because they were **conspicuous against the light lichen/background** and therefore were **easily spotted by the predators/birds**;

More of the dark form **survived, reproduced** and **pass on the allele for dark colour to their offsprings**.

Resulting in the **increase in allele frequency of dark colour**/ **dark forms predominate the population**.

(c) The life cycles of bacteriophages can be classified as 'lytic' or 'lysogenic'. These terms are sometimes also used for animal viruses.

Use your knowledge of 'lytic' and 'lysogenic' to contrast the lytic life cycle of the influenza virus with the lysogenic life cycle of HIV. [5]

- Define 'lytic' and 'lysogenic'
  Lytic – reproductive cycle that involves the lysis of the host cell.
  Lysogenic – reproductive cycle that does not involve the death of the host cell. Involves the integration of viral genome into the host genome. Viral genome is replicated as host genome is replicated ; [max 1m for Definition]

- Contrast lytic life cycle of influenza virus with lysogenic life cycle of HIV

SC	**OR**	**E**	
		Lytic (influenza)	**Lysogenic (HIV)**
DIFFERENCES	intent / Purpose	For the mass reproduction of viruses	For integration into host genome and to remain dormant until an appropriate environmental stimulus triggers mass reproduction.
DIFFERENCES	Gene expression	Genes coding for structural components such as capsid and enzymes such as RNA-dependent RNA polymerase are synthesized for the assembly of the virus.	gene which codes for reverse transcriptase and integrase are expressed to allow for the genome of HIV to be integrated into host genome.
DIFFERENCES	Reproducti on of virus	Newly-assembled phages formed are ready and will be released by the host cell.	No new viruses are formed. Instead, viral DNA replicates as part of bacterial DNA (in the form of prophage).
DIFFERENCES	Replicatio n of Viral Genome	RNA dependent RNA polymerase from virus is involved in making more copies of viral RNA.	Reverse transcriptase needed to convert HIV RNA into DNA, to be integrated into host genome. HIV genome replicated as host genomeis replicated.
DIFFERENCES	Fate of Host	Host cell dies upon release of new viruses.	Host cell survives and every daughter bacterium now contains a prophage.

[Total: 20]

10 (a) Describe the pathway taken by cell-membrane bound proteins following their synthesis. [7]

- Proteins enter lumen of rough endoplasmic reticulum;
- Embedded into phospholipid membrane of transport vesicles;
- Transport vesicles bud from the rough endoplasmic reticulum;
- Transport vesicles move to the Golgi apparatus along microtubules;
- Transport vesicles fuse with the *cis* face of the Golgi apparatus;
- Travel through cisternae of Golgi apparatus;
- Secretory vesicles bud from the *trans* face of the Golgi apparatus;
- Secretory vesicles travel to the cell-membrane;
- Secretory vesicles fuse with the cell-membrane;
- Proteins become incorporated into the cell-membrane; [max 7 marks]

(b) Describe the roles of NAD and NADP in cells. [8]

- Both NAD+ and NADP+ are **coenzymes** and **electron acceptors/carriers**. In reduced form, NADH and NADPH respectively, temporarily **store energized electrons and protons** which transfer them to electron transport chain (ETC) ;

- NAD which is found in the mitochondrial matrix and cytoplasm of cells, is **reduced** to NADH during glycolysis, link reaction and Krebs cycle by dehydrogenation reactions
- NADH **donates its electrons to the electron transport chain (ETC)** embedded in the **inner** mitochondrial membrane during **oxidative phosphorylation** ;
- The transfer of the electrons through ETC down the energy gradient results in the establishment of a proton concentration gradient and eventually **ATP synthesis** ;
- **Each NADH produces 3 ATP**. The regeneration of NAD+ ensures that glycolysis, link reaction and Krebs cycle of cellular respiration can continue to proceed for the synthesis of ATP via substrate level phosphorylation or oxidative phosphorylation ; [max 4 marks]

- NADP, found in the **stroma** of chloroplast in plant cells, is the **final electron acceptor** of the ETC embedded at the thylakoid membrane. The reduced form, **NADPH**, is produced during the **light dependent reaction** of photosynthesis. They are needed for the **light independent / Calvin cycle** ;
- **NADPH reduces glycerate-3-phosphate to form triose phosphate/glyceraldehyde-3-phosphate**. Some of the triose phosphates will combine to form other sugar phosphates and eventually produce starch, fats and proteins in plants ;
- By **donating its hydrogen** to glycerate-3-phosphate, **NADP+ is regenerated** and continues to accept the electrons and protons from the light dependent reaction for the synthesis of different bio-molecules in plants ;
- The rest of triose phosphates are used for **regeneration of ribulose bisphosphates (RuBP)** needed for **carbon dioxide fixation** ; [max 8 marks]

(c) Describe how the differences in the structure and organization of genetic material in eukaryotes and prokaryotes affects the way genes are expressed. [5]

Contrast in Structure/ Organisation of Genetic Material	Prokaryotes	Eukaryotes
Structure by modifying chromatin	1. Prokaryotic DNA not organised into chromatin and is easily accessible to RNA polymerase Prokaryotes have 2 levels of packing: formation of looped domains and DNA supercoiling	Compact chromatin fibre (characterised by nucleosome coiling) to be uncoiled to allow access of DNA to RNA polymerase II and transcription factors Eukaryotes have 4 levels of packing: formation of nucleosome (10nm fibre), 30nm fibre solenoid, 200nm looped domains and 1400 nm metaphase chromosomes
	2. Prokaryotic DNA does not form chromatin / not associated with histones	DNA modification such as DNA methylation and histone modification (e.g. histone acetylation) allow conversion between euchromatin and heterochromatin
Structures referred as a switch	3. DNA sequences, such as operators, serve as the on/off switch	Chromatin structure, i.e. euchromatin, (readily transcribed) vs heterochromatin(not available for transcription), is the major on/off switch
Regulatory sequences	4. Promoter (with Pribnow box) 5. Operator site of operon 6. Transcriptional regulatory protein / Regulator protein binds to DNA-binding sites upstream of the cluster of structural genes to regulate initiation of transcription.	Promoter (with TATA box) Promoter proximal elements Distal control elements which form enhancers and silencers More extensive interaction between upstream DNA sequences and protein factors involved to stimulate and initiate transcription. In addition to promoters, enhancers and silencers control rate of transcription initiation
Structures for inhibition of transcription	7. An operator site; lies between the promoter and the structural genes; A repressor binds to operator; to inhibit transcription;;	The promoter is located next to the gene it controls; A repressor binds to a distal control element called a silencer; and inhibits RNA polymerase from binding to the promoter thus inhibiting transcription;;
Structures which enhance transcription	8. The operon can be under positive control such as in the case of the lac operon where a CAP binding protein binding to the CAP binding site; located before the promoter enhances rate of transcription;	Activators bind to distal control elements known as enhancers ; and stabilize the binding of RNA polymerase to the transcription initiation complex; enhancing rate of transcription
Need for transcription initiation complex to be assembled at promoter	9. No general transcription factors are needed to bind to the promoter to create the transcription initiation complex for transcription;	General transcription factors are needed to bind to the promoter to create the transcription initiation complex for transcription;
Introns	10. Absence of introns → 1 gene codes for 1 polypeptide chain	Introns present where alternative splicing can occur resulting in 1 gene → 2 or more polypeptide chains
Arrangement of genes in the transcription unit	11. Multiple genes are located side by side in an operon and controlled by a single promoter	Each transcription unit consists of a gene is controlled by its own promoter.

[Max 5 marks]

[Total: 20]

END OF PAPER